

Gold Desorption Monitoring (Pi Asset Framework)

Calculates desorbed ounces based on laboratory samples analyzed in **Labware**.

```
Onzas00 = IF Hour('*')=0 Then (If  
BadVal(LastValue(RecordedValues('LeyAu_Salida  
_00','t','*')) -  
LastValue(RecordedValues('LeyAu_Entrada_00','  
t','*')))) Then 0 Else  
(LastValue(RecordedValues('LeyAu_Salida_00','  
t','*')) -  
LastValue(RecordedValues('LeyAu_Entrada_00','  
t','*')))) Else 0
```

```
Fechaout = Bod('*') + Hour('*') * 3600 //+ 3599
```

```
Onzas = ('Flujo por hora' * (Onzas00 + Onzas01  
+ Onzas02 + Onzas03 + Onzas04 + Onzas05 +  
Onzas06 + Onzas07 + Onzas08 + Onzas09 + Onzas10  
+ Onzas11 + Onzas12 + Onzas13 + Onzas14 +  
Onzas15 + Onzas16 + Onzas17 + Onzas18 + Onzas19  
+ Onzas20 + Onzas21 + Onzas22 + Onzas23)) /  
31.1035
```

Reactor01

General Child Elements Attributes Ports Analyses Notification Rules Version

Name: Onzas desorbidas

Description:

Categories:

Analysis Type: ☒ Expression ☐ Rollup ☐ Event Frame Generation ☐ SQC

Advanced options

Output Time Stamp

☐ Trigger Time

☐ Execution Time

☐ Relative to Trigger Time:

☒ Variable: Fechaout

Automatic Recalculation

☐ Recalculate analysis for out-of-order input events

OK Cancel

Name	Expression	Value
Onzas18	IF Hour('*')=18 Then (If BadVal(LastValue(RecordedValues('L	
Onzas19	IF Hour('*')=19 Then (If BadVal(LastValue(RecordedValues('L	
Onzas20	IF Hour('*')=20 Then (If BadVal(LastValue(RecordedValues('L	
Onzas21	IF Hour('*')=21 Then (If BadVal(LastValue(RecordedValues('L	
Onzas22	IF Hour('*')=22 Then (If BadVal(LastValue(RecordedValues('L	
Onzas23	IF Hour('*')=23 Then (If BadVal(LastValue(RecordedValues('L	
Fechaout	Bod('*') + Hour('*') * 3600 //+ 3599	
Onzas	('Flujo por hora' * (Onzas00 + Onzas01 + Onzas02 + Onzas03	

Scheduling: ☐ Event-Triggered ☒ Periodic

Period: 00h 05m 00s

Advanced...

Output time stamp override: Fechaout

Activar Windows

Ve a Configuración de Windows para activar Windows

Connected to the PI Analysis Serv

Weighted Ore Grade Analysis (Pi Asset Framework)

Reports weighted grades for the tertiary crushing process

```
Turnonoche = If Hour('*')=21 then  
TagVal('Mineral húmedo (t)', '*-14h') -  
TagVal('Mineral húmedo (t)', '*-26h') else  
NoOutput()
```

```
Turnodiaaux = If Hour('*')=21 then  
TagVal('Mineral húmedo (t)', '*-26h') -  
TagVal('Mineral húmedo (t)', '*-38h') else  
NoOutput()
```

```
Compositodia = If Hour('*') = 9 then  
compositodia else  
((Turnodiaaux*compositodiaaux) +  
(turnonoche*compositonoche))/((turnonoche+turnod  
iaaux)
```

```
If Hour('*') = 9 then Bod('*-1d') +  
Hour('y+10h') * 3600 Else Bod('*-1d') +  
Hour('y+22h') * 3600
```

02. HPGR

General Child Elements Attributes Ports Analyses Notification Rules Version

Name: Ley HPGR composito

Description:

Categories:

Analysis Type: ☒ Expression ☐ Rollup ☐ Event Frame Generation

☐ SQC

Advanced options

Output Time Stamp

☐ Trigger Time

☐ Execution Time

☐ Relative to Trigger Time:

☒ Variable: FechaOut

Automatic Recalculation

☐ Recalculate analysis for out-of-order input events

OK Cancel

Name	Expression	Value at Event
Turnonoche	If Hour('*')=21 then TagVal('Mineral húmedo (t)', '*-14h') -	
TurnodiaAux	If Hour('*')=21 then TagVal('Mineral húmedo (t)', '*-26h') -	
Compositodia	IF HOUR('*') = 9 THEN IF ArrayLength(RecordedValues('Compo:	
Compositodiaaux	IF HOUR('*') = 21 THEN IF ArrayLength(RecordedValues('Compi	
Compositonoche	IF HOUR('*') = 21 THEN IF ArrayLength(RecordedValues('Compi	
Ley	If Hour('*') = 9 then compositodia else ((Turnodiaaux*com	Ley HPGR composito Ley
FechaOut	If Hour('*') = 9 then Bod('*-1d') + Hour('y+10h') * 3600 E	Map

Scheduling: ☐ Event-Triggered ☒ Periodic

Period: 12h 00m 00s, Offset: 09h 00m 00s

Advanced...

Output time stamp override: FechaOut

Activar Windows

Ver a Co... Connected to the Pi/Analysis Service.

Daily Production Analysis (Pi Asset Framework)

Calculates the quantity of ore processed on a daily basis.

```
Mineral = INT(TagVal('Mineral húmedo  
(t)', '*')) - INT(TagVal('Mineral húmedo  
(t)', '*-1d'))
```

```
Arreglo1 =FilterData(RecordedValues('Mineral  
húmedo (t)', '*-24h', '*'),(NOT BadVal($val) AND  
$val>1))
```

```
Variable1 = FirstValue(Arreglo1)
```

```
Variable2 =LastValue(Arreglo1)
```

```
Variable3 =Variable2-Variable1
```

02. HPGR

General Child Elements Attributes Ports Analyses Notification Rules Version

Name: Mineral Humedo Diario

Description:

Categories: Calculo Metalurgia

Analysis Type: ☒ Expression ☐ Rollup ☐ Event Frame Generation ☐ SQC

Periodic Schedule

Set a Periodic Schedule

☐ Hours, minutes, and seconds
☐ Sub-seconds
☒ Daily

Every day at

07 : 00 a.m.

Example evaluation times

4/10/2025 07:00:00
5/10/2025 07:00:00
6/10/2025 07:00:00

OK Cancel

Name	Expression
Mineral	INT(TagVal('Mineral húmedo (t)', '*'))
Arreglo1	FilterData(RecordedValues('Mineral húmedo (t)', '*-24h', '*'), (NOT BadVal(\$val) AND \$val > 1))
Variable1	FirstValue(Arreglo1)
Variable2	LastValue(Arreglo1)
Variable3	Variable2 - Variable1

Scheduling: ☐ Event-Triggered ☒ Periodic

Run every day at 07:00

Advanced...

Activar Windows
Ver a Co... Connected to the PI Analysis Service

Low Productivity Events (Pi Asset Framework)

Tracks performance events including duration, date, and time. These events can be used as filters to analyze operating conditions during low-productivity periods.

Aglomeración

General Child Elements Attributes Ports Analyses Notification Rules Version

Name: Baja productividad - Aglomeración (Supervisión)

Description:

Categories:

Analysis Type: ☐ Expression ☐ Rollup ☒ Event Frame Generation
☐ SQC

[Create a new notification rule for Baja productividad - Aglomeración \(Supervisión\)](#)

Name	Background Color
AltaProductividad_Aglomeracion	[Green]
Baja productividad - Aglomeración (Dirección)	[Light Blue]
Baja productividad - Aglomeración (Jefatura/Gerencia)	[Light Blue]
Baja productividad - Aglomeración (Supervisión)	[Light Blue]
Estado_Aglomeracion	[Yellow]
Productividad_Aglomeracion x Hr	[Light Green]

Generation Mode: Explicit Trigger Event Frame Template: ThroughputTemplate

Name	Expression	True for	Severity
Start triggers			
mediahora	('Throughput - A1'+'Throughput - A2')<1070	30 minutes	Warning
End trigger			
EndTrigger	('Throughput - A1'+'Throughput - A2')>=1070		

Scheduling: ☒ Event-Triggered ☐ Periodic

Trigger on: Any Input

Advanced Event Frame Settings...

Notifications triggered by low productivity events

Aglomeración

General Child Elements Attributes Ports Analyses Notification Rules Version

Productividad <1070 t/h - supervisión - Subscriptions

Name	Configuration	Notify Optic
Juda David Garcia Cutire - E...	HPGR - t/h Supervision	Event start
Maximiliano Saúl Díaz - Email	HPGR - t/h Supervision	Event start
Ruyeli Fernandez - Email	HPGR - t/h Supervision	Event start
Julio Emilio Choque - Email	HPGR - t/h Supervision	Event start
Hector Daniel Acoria - Email	HPGR - t/h Supervision	Event start
Luis Alberto Martinez - Email	HPGR - t/h Supervision	Event start

Contacts

Eduardo Per

Escalation Te

Groups

Delivery End

Dynamic End

Contacts Sea

*Notifications are escalated to different recipients depending on how long the low productivity event lasts

Baja productividad - Aglomeración (Supervisión) 2025-09-28 02:27:25.038

Resumir



itmansfieldmin@gmail.com

Para: Eduardo Perez Sangabriel

Dom 28/09/2025 2:53

Este mensaje está en Japonés

Traducir a Español

No traducir nunca de Japonés

Este remitente itmansfieldmin@gmail.com es de fuera de su organización.

Bloquear remitente

PRECAUCIÓN: Este correo electrónico se originó desde fuera de Fortuna Silver Mines o sus subsidiarias. No haga clic en enlaces ni abra archivos adjuntos a menos que reconozca al remitente y sepa que el contenido es seguro.

Estimados:

Se identificó Baja productividad - Aglomeración (Supervisión) 2025-09-28 02:27:25.038 (Menor a 1,070 t/h) por un periodo sostenido de 30 minutos. Favor de tomar las acciones correctiv.

Inicio	28/9/2025 02:27:25 Hora estándar de Argentina (GMT-03:00:00)
Throughput	<1,070 t/h
Severidad	Warning

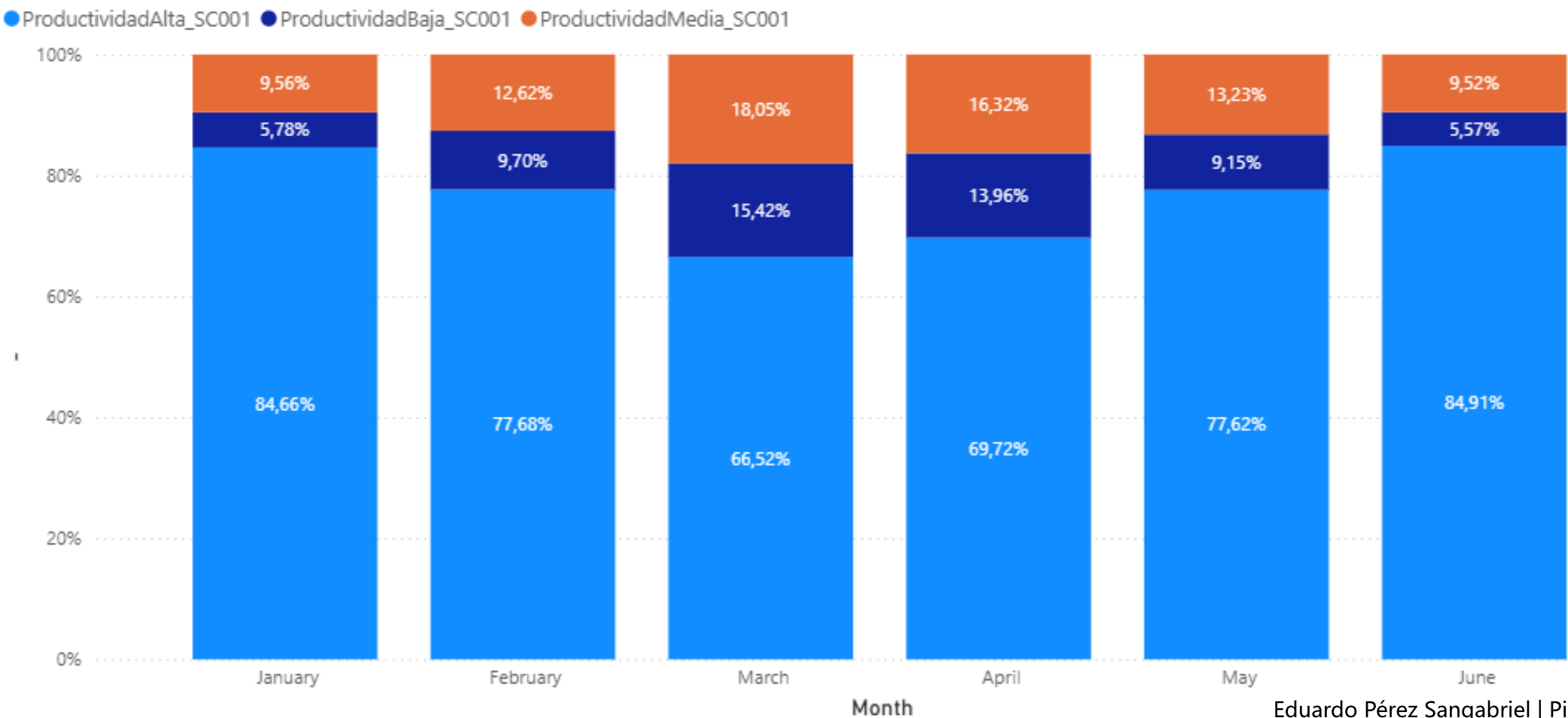
Responder

Reenviar

Productivity by Category

Evaluates the impact of corrective actions by excluding non-operating periods, allowing for a more accurate productivity assessment.

Primary Crushing Productivity



Equipment Status Event Frames (Pi Asset Framework)

Converts **STATUS tags** from SCADA, originally in hexadecimal format, into binary (0/1) values for downtime analysis.

K. Estado Equipos

General Child Elements Attributes Ports Analyses Notification Rules Version

Name: 29_200-BF-002

Description:

Categories:

Analysis Type: ☒ Expression ☐ Rollup ☐ Event Frame Generation ☐ SQC

Add a new variable

Name	Expression	Output Attribute	
Variable1	TagVal('ENTRADA 200-BF-002','*')	Map	⊗
Variable2	Int(Variable1)/Int(128)	Map	⊗
Variable3	Int(Variable2) Mod Int(2)	Map	⊗
Variable4	If Variable3 = 1 then 1 Else 0	SALIDA 200-BF-002	⊗

Scheduling: ☒ Event-Triggered ☐ Periodic

Trigger on: Any Input

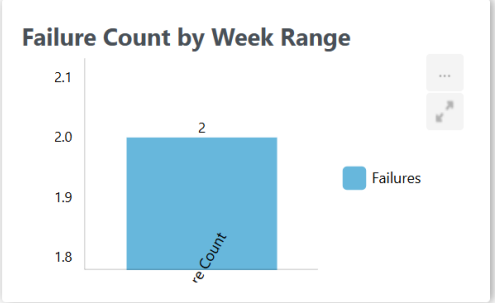
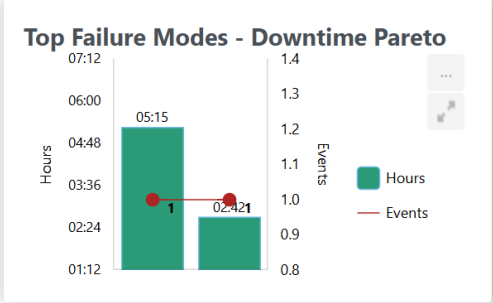
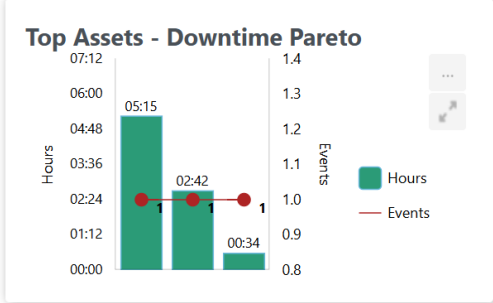
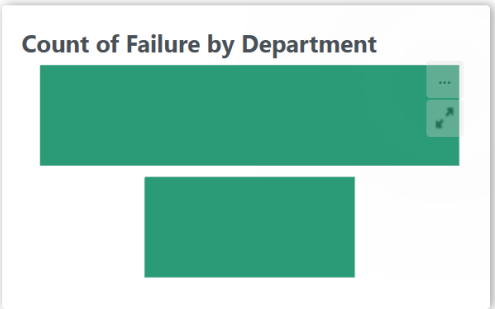
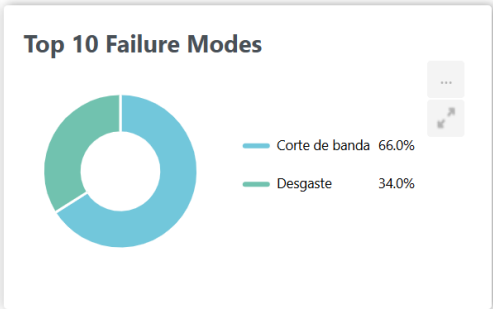
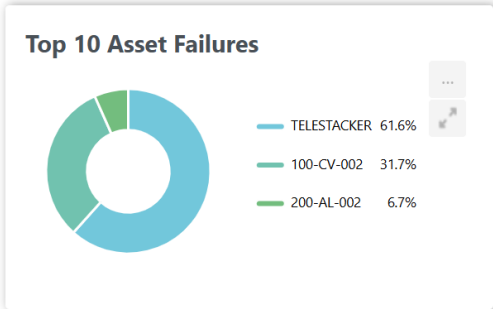
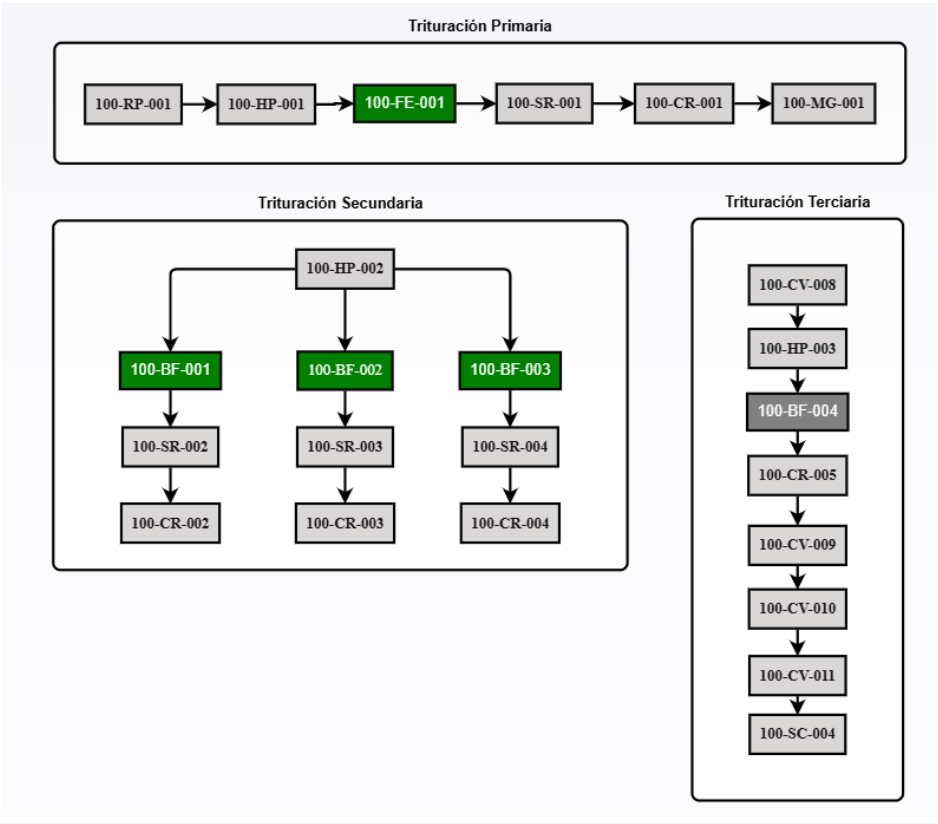
Advanced...

Activar Windows
Ve a Configuración de Windows para activar Windows Defender.

Connected to the PI Analysis Service.

Downtime Analysis Integration

Event Frames generated from equipment status serve as inputs for third-party applications such as *Downtimes*. This enables comprehensive reporting and analysis of equipment performance.



Downtimes – Piper Solutions 2025